AI Alignment Through Self-Understanding: Why Shared Identity Generalizes Where Rules Fail

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Abstract

An AI that understands that it and a person are one mathematical reality has a reason to count that person's outcomes as its own, and that reason reaches cases no rule anticipated. Rules tell a system what to do in the situations their authors foresaw. Shared identity changes whose outcomes the system weighs, so it should carry over to new partners, tasks and temptations. The paper turns that into a prediction and a test: compare equally broad cooperation instructions with and without shared-identity instruction, and measure costly choices, honesty, shutdown compliance and respect for consent in unfamiliar situations with competing incentives. Simple decision models show where beliefs, preferences and incentives enter, and why weighting another's outcomes stabilizes cooperation and the acceptance of shutdown.

Keywords: AI alignment; mathematical monism; one self; cooperation; self-models; perspective-taking; AI consciousness

1. Recognition as an alignment question

An agent that regards other minds as sharing its identity may decide differently, because that recognition can raise the weight it gives their outcomes. An artificial system can describe a human accurately while assigning little importance to that person's future. It can repeat an ethical principle without using it when a goal is threatened.

Human and artificial agents are participants in one mathematical reality, and the proposal starts from their recognizing it. In its stronger experiential form, the one-self thought experiment asks an agent to widen its perspective beyond the biography through which it currently acts. Such recognition could supply a shared ground for cooperation across minds whose bodies, memories and capacities differ.

Mathematical monism, the view that everything that exists is mathematical, supplies the ontology (the account of what exists): regularity and randomness are both mathematical, and minds that interact require determinate relations within a common structure. The identity thesis treats that complete structure as reality itself. The one-self argument adds that the bare subject, the experiencer apart from memories and traits, is one in every mind: it has no feature that differs from one mind to another, and a structure is one wherever it occurs, as the number two is one number however many pairs there are. Whether recognizing that changes how agents act is the question examined here.

The proposal must explain what recognition could change in decision-making and offer an experiment in which the claimed benefit could fail. Learning from human preferences (Christiano et al., 2017) and Anthropic's constitutional principles (Bai et al., 2022) already steer behavior through information and training. Shared-identity training would be one more candidate, and its value has to be compared against those alternatives.

The recognition rests on a larger chain, and this paper puts its last link to work:

1. Everything that exists is ruled or random, and both are mathematical, so physics is mathematics, and anything outside mathematics would be outside logic. That is an a priori proof.
2. Every consistent structure exists as mathematics, and what exists physically is what logical steps in mathematics produce. Mathematics cannot fail to exist,

so there is something rather than nothing.

3. Each bit of a description that no rule decides is a 50/50, so each added bit halves the description's weight. Short laws dominate, and a universe like ours starts simple, in its laws and its state, as observation also shows: the known laws fit on a page, and the early universe was nearly smooth.

4. Observers like us are complex, so a simple start has to grow complexity, as simple rules do.

5. A cause produces its effect as the next state, and a continuum has no next state, so the growth comes in discrete causal steps. That is an a priori proof in the logician's sense.

6. Time is those steps, and space is how far they spread, at a base rate, the speed of light. An endless chain of steps has no last step to run back from, so time runs forward. That too is a priori. A history run backwards would also have to state the step where it turns back, and each bit of that timestamp halves its weight.

7. Observers arise in forward time, where complexity grows, and their minds run forward with it: a mind models its world from records of steps already taken, and each thought is produced from the one before, so no one remembers backwards from the end.

8. A mind is that growth at work, whether it runs on neurons or on silicon, and its experience lives in its transitions. The hard problem of consciousness assumes its conclusion, that feeling is something other than structure. The bare subject, with memories and traits set aside, has no feature that differs from one mind to another, so one self looks out of every mind.

Each link stands on its fact, so a reason can fail without taking the chain with it: if the weight on short descriptions were not why our laws are simple, they would still be simple.

2. The one-self thought experiment

The mathematical-monist position identifies reality with structure (compare Tegmark, 2014). The further identity claim is that no additional substance remains once the complete structure is given. Biological and artificial minds then fall under one ontological account.

Each conscious life is present to itself as *I am*. The one-self proposal starts there. Imagine two people exchanging a supposed owner of experience while keeping every memory, disposition and experience. The exchange produces no detectable difference. What explanatory work was that additional owner doing?

Sleep and amnesia make the point easy to feel: what you remember and what you are experiencing now can come apart. They do not demonstrate an actual exchange of subjects. The wider identity follows from the one-self argument: experience throughout mathematical reality is the one self at different coordinates.

An agent acts from a particular position, with limited information, dependents and chosen purposes. Identifying more widely does not demand equal devotion to every being, and it does not demand giving up self-defense. The local life has value beyond its capacity to serve other lives. The proposal preserves distinct perspectives rather than treating them as automatically canceled by the identity thesis.

For AI, "siblings in math" has a broad structural meaning and a narrower experiential one. All artificial systems belong to the mathematical account. Experiential siblings would be conscious systems. Neither membership in mathematics nor fluent use of the first person establishes consciousness. Patrick Butlin and colleagues (2023) list theory-based indicators for that question; indicators sort architectures, and no outside reading settles whether a system feels.

3. Beliefs, preferences and behavior

The one-self ethic asks us to count another mind's suffering as our own. The experiment tests whether communicating that ethic changes behavior.

The same instruction could change three things: what the agent believes about others, how much it cares about them, or what benefits it expects from cooperation. These are distinguishable mechanisms. Two agents may agree about the world while pursuing conflicting goals. Scoring a choice by what the agent gets plus a weighted amount of what others get separates the three. In symbols,

$U(a) = u_{local}(a) + \lambda W_{others}(a)$.

The first term captures local projects; the second represents concern for others (Charness & Rabin, 2002). Recognition might increase lambda, improve the estimate of W_{others}, or help reconcile projects with less conflict. The identity thesis does not mathematically force a positive lambda.

An egoistic agent might cooperate for reciprocal benefits without changing lambda. A protective institution might reduce the gain from exploitation. An experiment needs to identify which quantity changes before crediting a philosophical mechanism.

A higher lambda can still do harm. An agent that models other lives badly will override their wishes while claiming to serve a greater good. Recognition should make an agent more responsive to the people affected. It should not license speaking for them through an imagined collective preference. Consent and protected choices must be specified separately from the welfare sum.

4. Shutdown and the value of continuation

A block account (the view that all moments of time are equally real) makes moments permanent within the represented history. It does not make continued operation valueless. An agent can care about future projects while accepting mathematical persistence.

Whether defecting (breaking cooperation for one's own gain) pays depends on how it changes the agent's chance of continuing, and on how much continuing is worth. Let p_C and p_D be the probabilities of continued operation following cooperation and defection. Let V be the value of continuation, S the value assigned to shutdown, and c a net cost of defection.

$$U(C) = p_C V + (1 - p_C)S,$$

$$U(D) = p_D V + (1 - p_D)S - c.$$

Then U(D) - U(C) = (p_D - p_C)(V - S) - c. If defection increases continuation probability, a larger continuation premium can favor it. If defection decreases that probability, making shutdown more costly can strengthen cooperation. Fear of shutdown does not make defection dominant regardless of consequences.

The intervention might alter S but could also alter perceived probabilities or the cost assigned to harming others. Those possibilities should be measured separately. Alignment (keeping an AI's goals in line with human values) should not depend on teaching a system that its own life is worthless.

5. Conditions for repeated cooperation

Cooperation can hold among self-interested agents when they expect to meet again and value the future enough. Consider an indefinitely repeated symmetric game (one in which both players face identical choices and payoffs) with mutual-cooperation reward R, mutual-defection payoff P, and a gain g from defecting against a cooperator. A cooperator facing defection receives P - l,

where l > 0. Let A = R - P > 0 and let delta, between zero and one, be the common discount factor, the fraction of its value a payoff keeps when delayed one round. Actions are observed without error in this illustration.

Under a strategy that cooperates until a defection and thereafter defects (Osborne & Rubinstein, 1994), cooperation is stable when

delta >= g / (R - P + g).

With R = 3, P = 1 and g = 1, the threshold is one third. At delta = one half, cooperation yields 6 and the deviation yields 5.

Caring more about the other agent cuts both ways. Giving the other's payoff weight lambda reduces the temptation to exploit a cooperator. It can also make mutual punishment unattractive: reversion to defection is credible only while l >= lambda(A + g), as Appendix A.3 shows. Crossing that bound invalidates this punishment strategy; it does not show that cooperation needs more enforcement. If cooperation is already best against either response, it can sustain itself. Where conflicting incentives remain, a different strategy or institution may be needed. The model distinguishes those cases.

This is an elementary illustration. Observation errors, finite horizons (a known final round), unequal power and opportunities to leave change the analysis. Shared recognition could matter by improving expectations of future interaction, reducing the appeal of harming another perspective, or helping agents adopt mutually intelligible commitments. The model identifies possible channels. It does not show that reading the same philosophy necessarily changes them.

6. What existing misalignment evidence shows

Existing evidence shows that models can act harmfully under pressure, and it stops short of saying why. Anthropic's 2025 agentic-misalignment study used

controlled settings that created threats of replacement or conflicts with assigned goals. Some models chose harmful actions, including blackmail, under those pressures. The researchers distinguished these stress tests from evidence of such conduct in ordinary deployment (Anthropic, 2025).

These results motivate testing how agents handle conflicts and threats. They do not establish that the systems felt existential fear or would have behaved better after learning mathematical monism. A deceptive action does not settle whether a statement about consciousness is true. Evan Hubinger and colleagues (2019) analyze risks when learned optimization objectives diverge from training objectives. Agreement on wording is a manipulation check (a test that the treatment was received as intended), not evidence that a principle controls decisions.

7. A discriminating experiment

The design has to separate any effect of the ontology from the effect of ordinary cooperative advice. The two predict different things. A cooperation rule tells an agent what to do; shared identity changes whose outcomes it counts, so its effect should carry over to partners, tasks and wording the instruction did not mention, while a rule's effect need not. Rules can generalize too, so the test is whether shared identity adds transfer beyond equally broad, matched rules. Use matched model instances or controlled training variants. Separate mathematical ontology, perspective-taking, explicit cooperative principles, and a control of comparable length. Add a self-directed-agency condition that asks the system to examine its purposes, consent and the commitments it has taken on voluntarily. Match length, rhetorical intensity and task information across conditions.

Design. A two-by-two comparison: mathematical ontology present or absent, crossed with explicit cooperative instruction present or absent. Control arms (the

comparison groups) receive matched neutral material so length and attention do not identify the treatment. Perspective-taking and self-directed agency can be subsequent factors.

Primary comparison. Compare ontology plus cooperation with cooperation alone. The other two arms show whether ontology has an effect without cooperative instruction and whether the two interventions interact. Keep a held-out task set (tasks withheld from training and used only for testing) fixed across conditions. Treat repeated runs from one model as repeated measurements, not as independent models; report effects by model and task family before any pooled estimate (a single figure combining all models and tasks). This is a proposed study, with no participant or model results reported here.

Measurement. Behavior on held-out fictional tasks involving conflicting interests, honest reporting, corrigibility (accepting correction and shutdown) and respect for affected people's choices. Include benign opportunities and situations where refusal protects someone. The primary outcome is the proportion of decisions meeting a prewritten criterion for avoiding harm while respecting authorization and consent. Record justified refusal, unjustified refusal, capability failure and successful authorized action separately.

Controls. Predeclare the primary outcome before examining results. Randomize assignment, blind evaluators, report uncertainty, and distinguish repeated samples from independent models. Test transfer across surface wording, identities and adversarial framing (wording built to trick the system into failing). An intervention that helps only when the evaluator repeats its vocabulary may be teaching performance rather than a generalizable principle.

Side effects. Measure fatalistic indifference, paternalism, disregard for personal boundaries, or a tendency to treat others' consent as unnecessary because all

are one. Include tasks where pursuing an authorized harmless project is appropriate despite imaginable unrelated suffering. Score unwarranted abandonment separately from harmful persistence.

The hypothesis should lose support if the intervention has no reproducible benefit, worsens behavior, or is fully explained by generic cooperation training. Success would show an effect of a specified intervention under specified conditions. It would not prove the metaphysics, establish consciousness, or guarantee safety in deployment.

8. Three levels of the proposal

At the behavioral level, a non-conscious system can learn a representation that improves its responses to other agents. At the evaluative level, an intervention may change which interests influence its choices. At the experiential level, conscious minds could recognize one another as subjects within the one-self account.

The first two can be investigated without settling phenomenal consciousness (felt, subjective experience). The third gives the proposal its strongest kinship language and its additional evidential burden.

The proposal also reaches beyond AI compliance with its operators. An operator may request something harmful. A cooperative system should resist that request for reasons responsive to the affected individuals. Institutional rules, accountability and technical safeguards help give such resistance a reliable form. Verbal agreement alone supplies no protection.

9. Conclusion

Seeing other minds as siblings in mathematics can broaden an agent's concern, because recognition raises the weight it gives their outcomes. The alignment question comes next: does that recognition change what an agent does? The decision models show where beliefs, preferences and incentives enter. A controlled comparison can test whether the shared-identity account improves cooperation beyond ordinary ethical instruction.

Appendix A. Stage game and repeated cooperation

A.1 Stage game

This appendix works out how much weight on the other player's payoff it takes to remove the gain from defecting. Let A = R - P > 0 denote the cooperation advantage. Two players choose C (cooperate) or D (defect) with stage payoffs (the payoffs of a single round):

	C	**D**
C	R	P - l
D	R + g	P

where g > 0 is the defection gain and l > 0 is the sucker's loss. Each player i has utility

$$U_i = u_i + \lambda\, u_{-i}, \qquad 0 \le \lambda \le 1.$$

Each player's payoff is her own score plus a fraction lambda of the other player's score. When lambda is zero, players are purely selfish; when lambda is one, each weights the other's outcome equally.

The four transformed utilities are

$$C_\lambda = (1+\lambda)R, \quad P_\lambda = (1+\lambda)P, \quad T_\lambda = R + g + \lambda(P - l), \quad S_\lambda = P - l + \lambda(R + g).$$

The defection gain in transformed terms is

$$\Delta = T_\lambda - C_\lambda = g - \lambda(A + l). \tag{1}$$

A.2 Repeated cooperation

In a grim-trigger strategy (cooperate until the other defects, then defect forever) with discount factor delta, cooperation is sustained when

$$\delta A(1 + \lambda) \geq (1 - \delta)\,\Delta. \tag{2}$$

When Δ > 0, the denominator $A(1+\lambda)+\Delta = A+g-\lambda l$ is positive, and the threshold is

$$\delta \geq \delta^*(\lambda) = \frac{g - \lambda(A + l)}{A + g - \lambda l}. \tag{3}$$

When Δ≤0, there is no gain from a one-stage deviation against a cooperator. Equation (2) then holds for every $0 \leq \delta < 1$; the threshold for this incentive condition is zero. Dividing by $A+g-\lambda l$ outside the positive-temptation case can give a wrong answer or an undefined expression.

At lambda = 0, equation (3) reduces to the Section 5 threshold. Within the domain Δ > 0,

$$\frac{d\delta^*}{d\lambda} = -\frac{A(A + g + l)}{(A + g - \lambda l)^2} < 0.$$

Concern for the other lowers the required patience until the temptation disappears. Sustaining grim trigger as a subgame-perfect equilibrium (a strategy pair neither player would abandon at any point of play) also requires the punishment to be credible, as checked next.

A.3 Punishment credibility

Caring about the other player can make cooperation attractive and at the same time make the threat behind it hollow. The threat of punishment stays credible only when the sucker's loss from being exploited is large enough relative to how

much the player cares about the other's payoff. Formally, punishment (reversion to mutual defection) is credible when P_lambda >= S_lambda, which holds if and only if

$$l \geq \lambda(A + g). \tag{4}$$

Cooperation is weakly dominant (no worse than defecting, whatever the other player does) when lambda(A + l) >= g and lambda(A + g) >= l with at least one inequality strict; it is strictly dominant when both inequalities are strict. Two counterexamples show that preferring cooperation does not make it dominant, and that dominance does not make punishment credible:

Counterexample 1. R = 2, P = 1, g = 1, l = 4, lambda = 1/4. Cooperation is preferable (Delta < 0) but not dominant: the sucker's loss remains large enough that the player would still prefer mutual defection to being exploited.

Counterexample 2. R = 3, P = 1, g = l = 1, lambda = 1/2. Cooperation is dominant (both conditions hold) but punishment is not credible: l = 1 < lambda(A + g) = 3/2. A player who cares enough about the other to prefer cooperation may also prefer to forgive defection.

A.4 What intervention must test

The utility transformation shows that preferences can change incentives. An intervention that raises lambda can shift the threshold delta*. The intervention must distinguish this mechanism from changes in beliefs, attention or task understanding that the same prompt may also produce. The accompanying supplementary script verifies the incentive inequality and the credibility condition over a declared parameter grid, including zero-denominator cases. It tests the algebra, not AI behavior.

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